



Rail Transit Track Structures & Components

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B O S T O N

Outline

- Basics of Transit Track Structure & Components
- Horizontal Curve, Superelevation, Spirals & Track Design
- Drainage
- Track Types
 - Ballasted Track
 - Embedded Track
 - Direct Fixation Track
- Guard Rails
- Fastening Systems
- Special Trackwork
- Misc. Track Items

Terminology

North American Nomenclature	English Nomenclature
Superelevation	Cant
Double Slip Switch	Double Slip/ Double Slip Point
Grade Crossing	Level Crossing
Spiral	Transition Curve
Turnout	Points
Switch	Points
Tie	Sleeper
Frog	Common Crossing
Grade	Gradient



Track Structures and Components

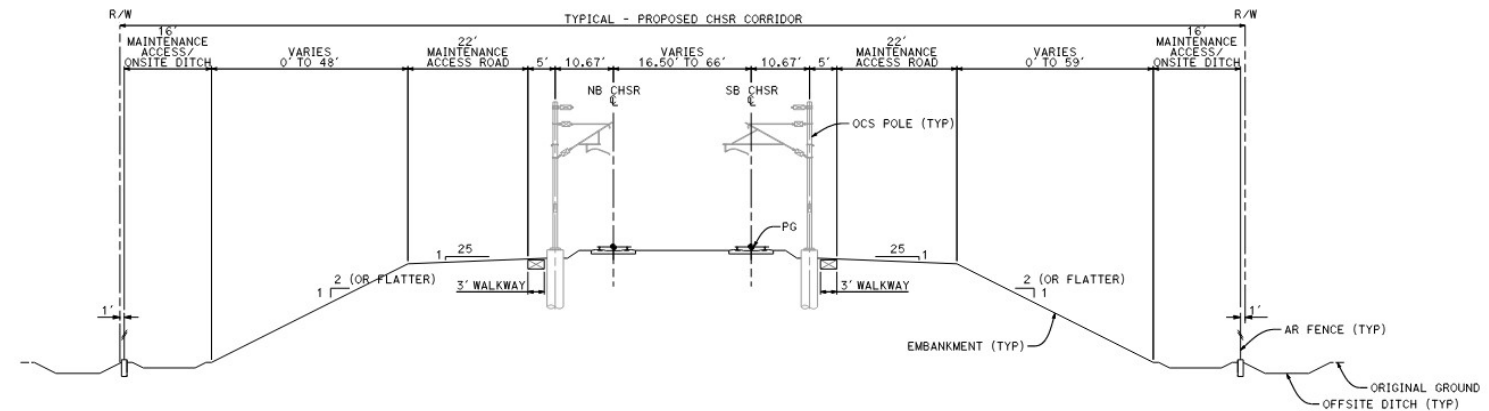
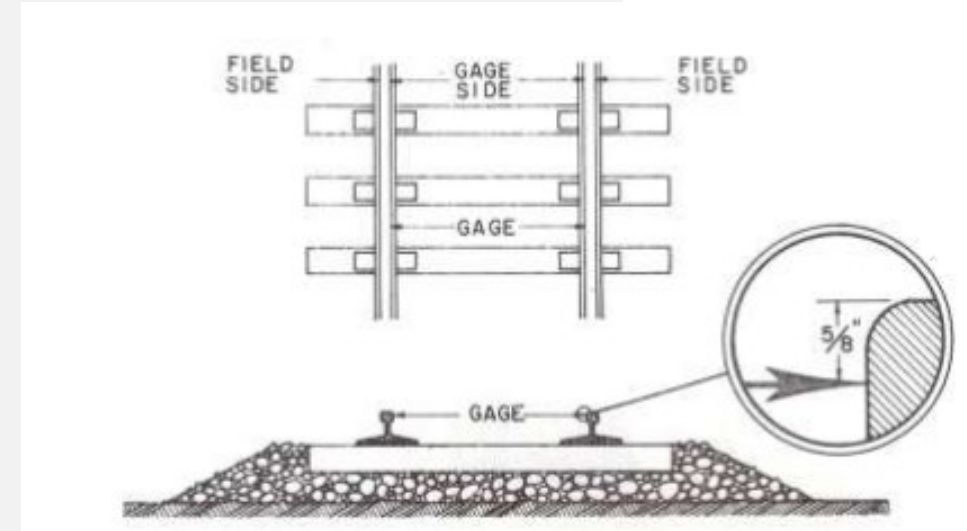
Ballasted Track

Direct Fixation Track

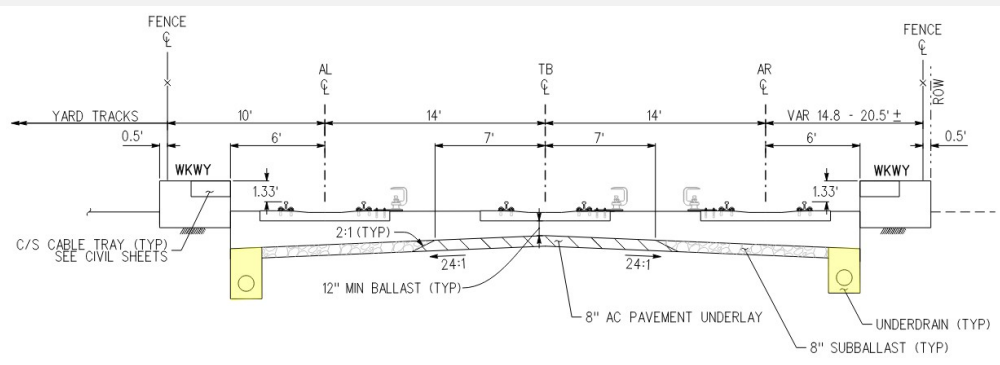
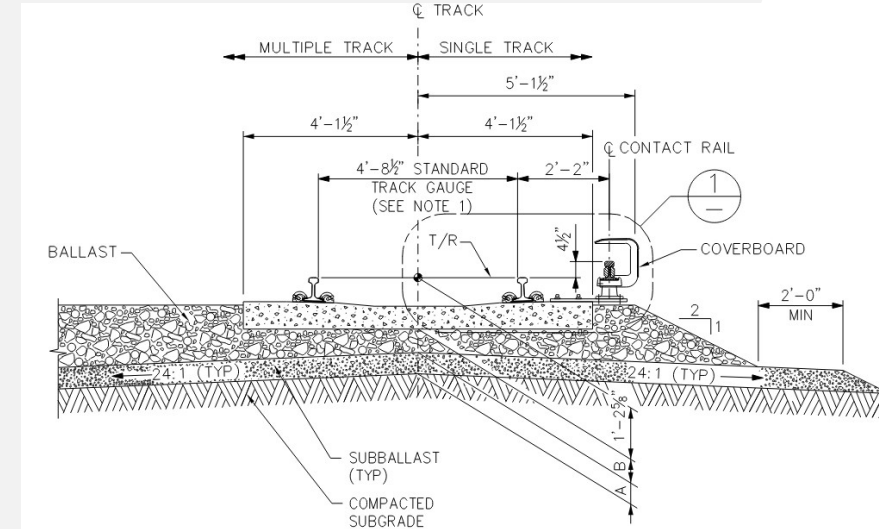
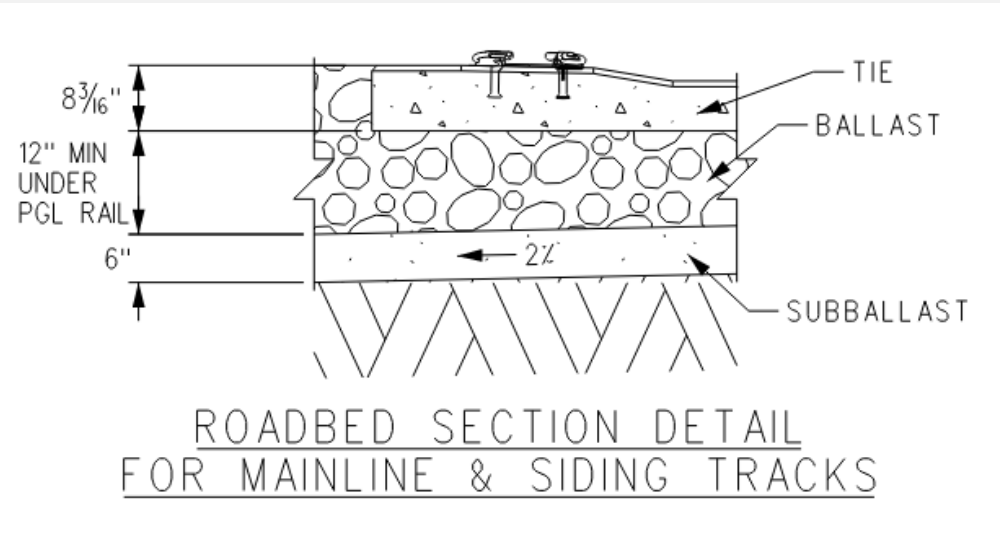
Embedded Track

Typical Sections

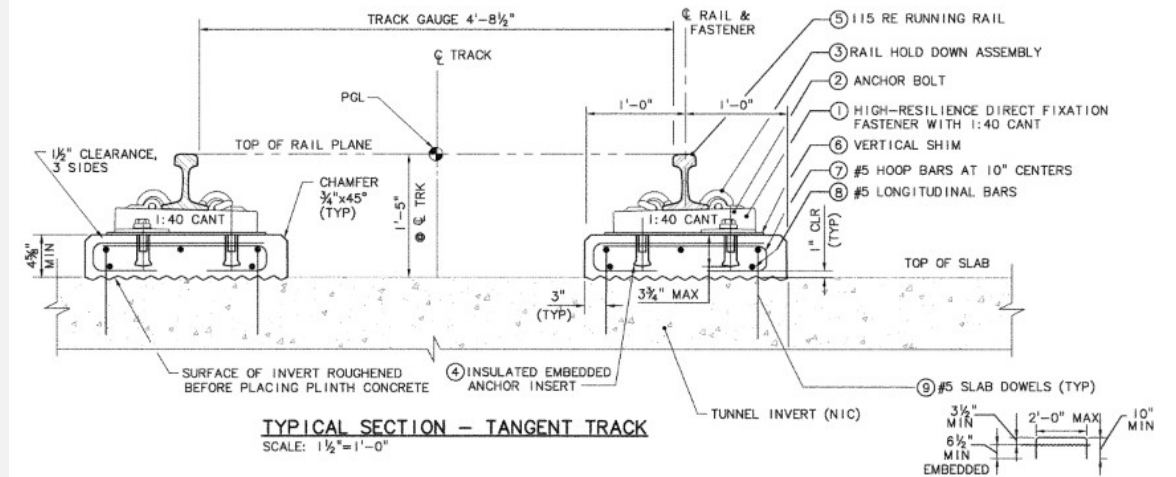
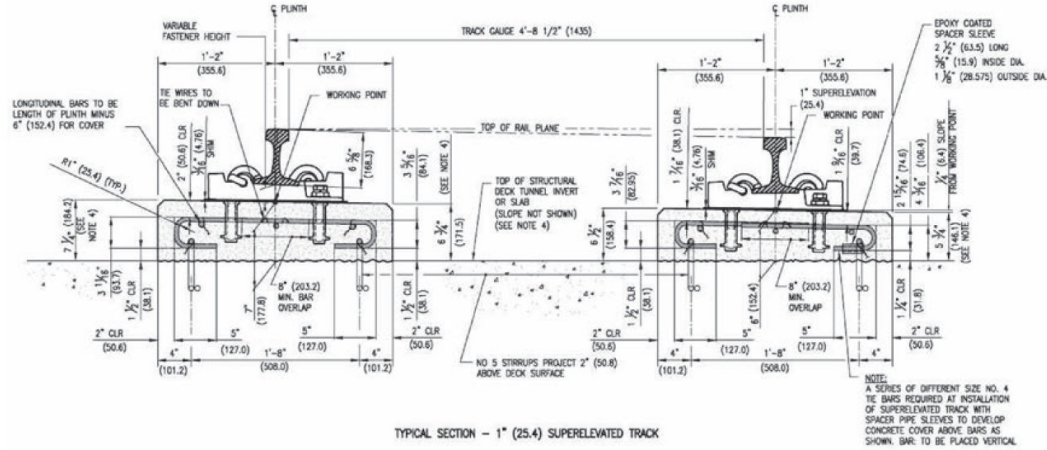
- Track centerline minimums (track centers)
 - Light Rail – 14'
- Gage – Distance between gage face of running rails
 - Standard gage 4'-8 1/2"
- Bedding material sloped to drain
- Wider shoulders for maintenance
- Drainage ditch parallel to tracks



Typical Sections – Ballasted Track



Typical Sections – Direct Fixation Track

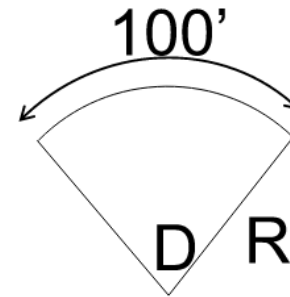
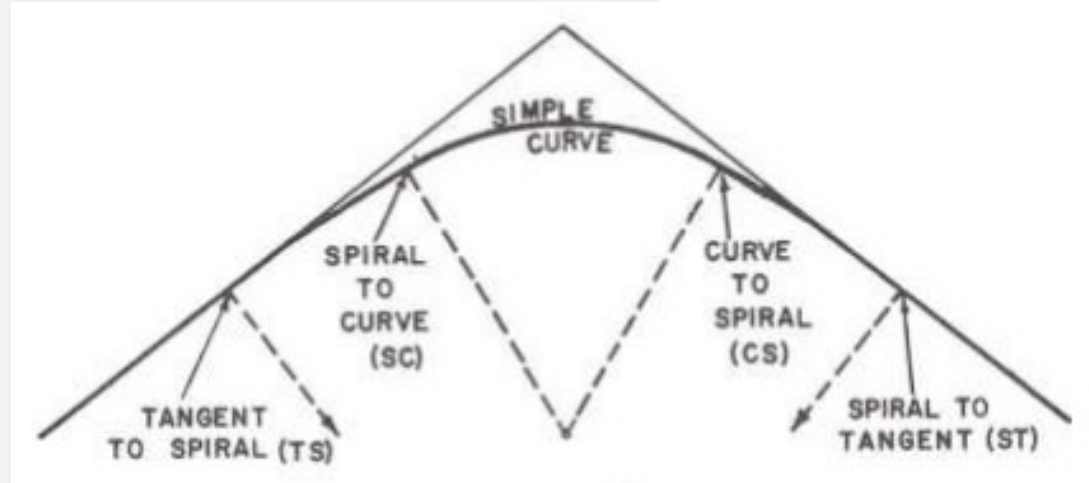




Horizontal Curve, Superelevation, Spirals & Track Design

Horizontal Design

- Goal is to have Highest Possible Speed
- Spiral Transitions
 - Eases train into a curve
 - Provides superelevation
- Rail Transit (Arc Definition)
 - Circular curves defined by arc and specified in radii.
 - Angle measured along the length of a section of curve subtended by a 100' arc

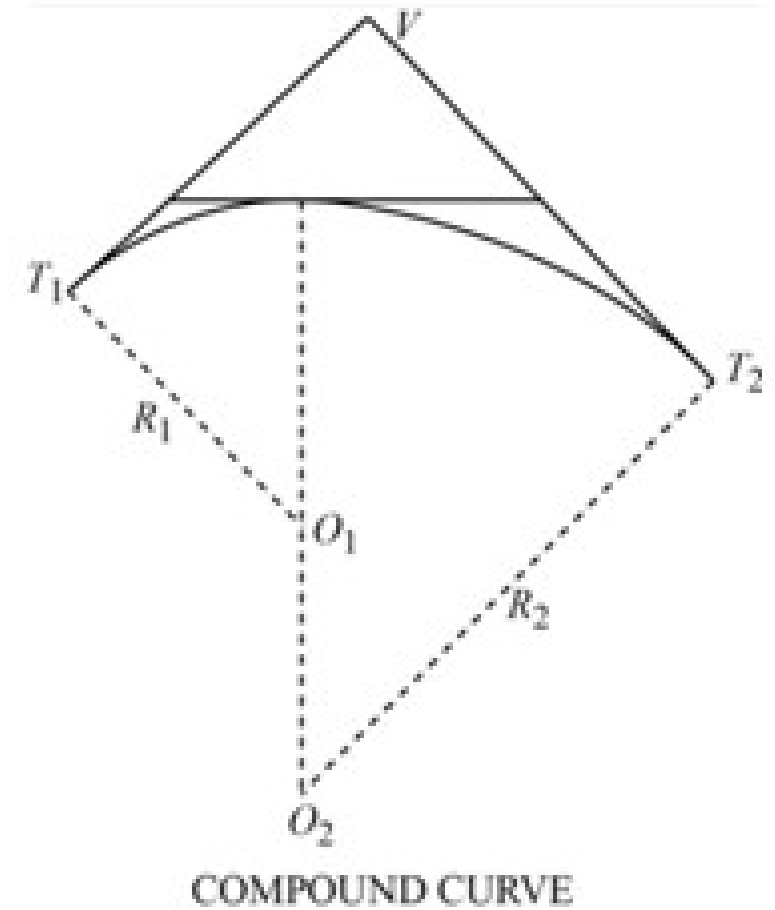
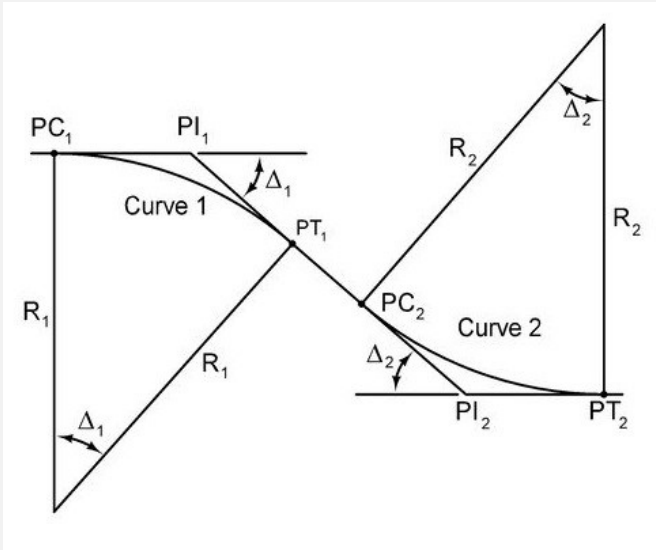


$$D/360 = 100/2(\pi)R$$

- 1-deg curve, $R = 5729.58'$

Horizontal Design

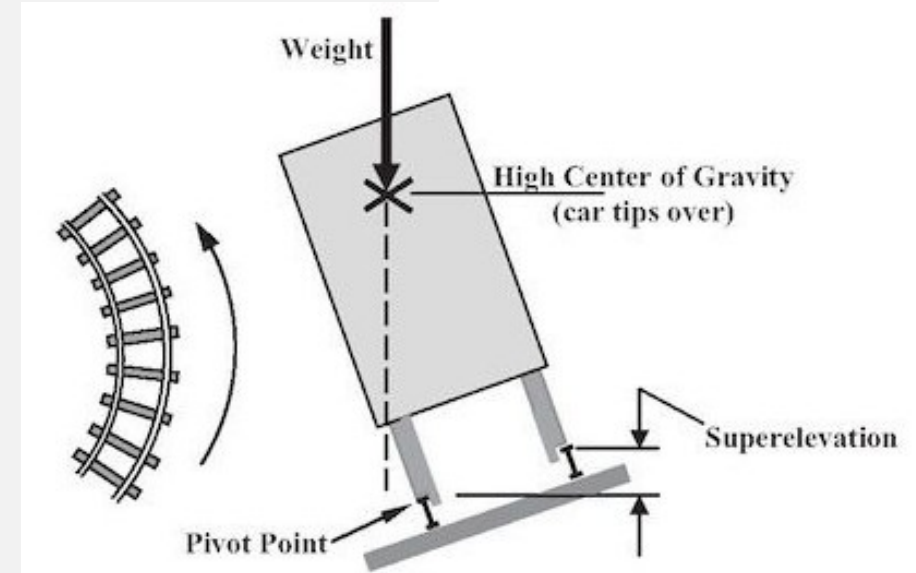
- Reverse Curves
 - Not preferred because it requires the speed on the track to be lower to account for the change in forces – if they are used, provide tangent for attenuation between curves
- Compound Curves
 - Two simple curves in the same direction separated by a transition curve. If difference in radii of curves is large, a spiral is needed between the curves



Superelevation

• Railway Superelevation, E

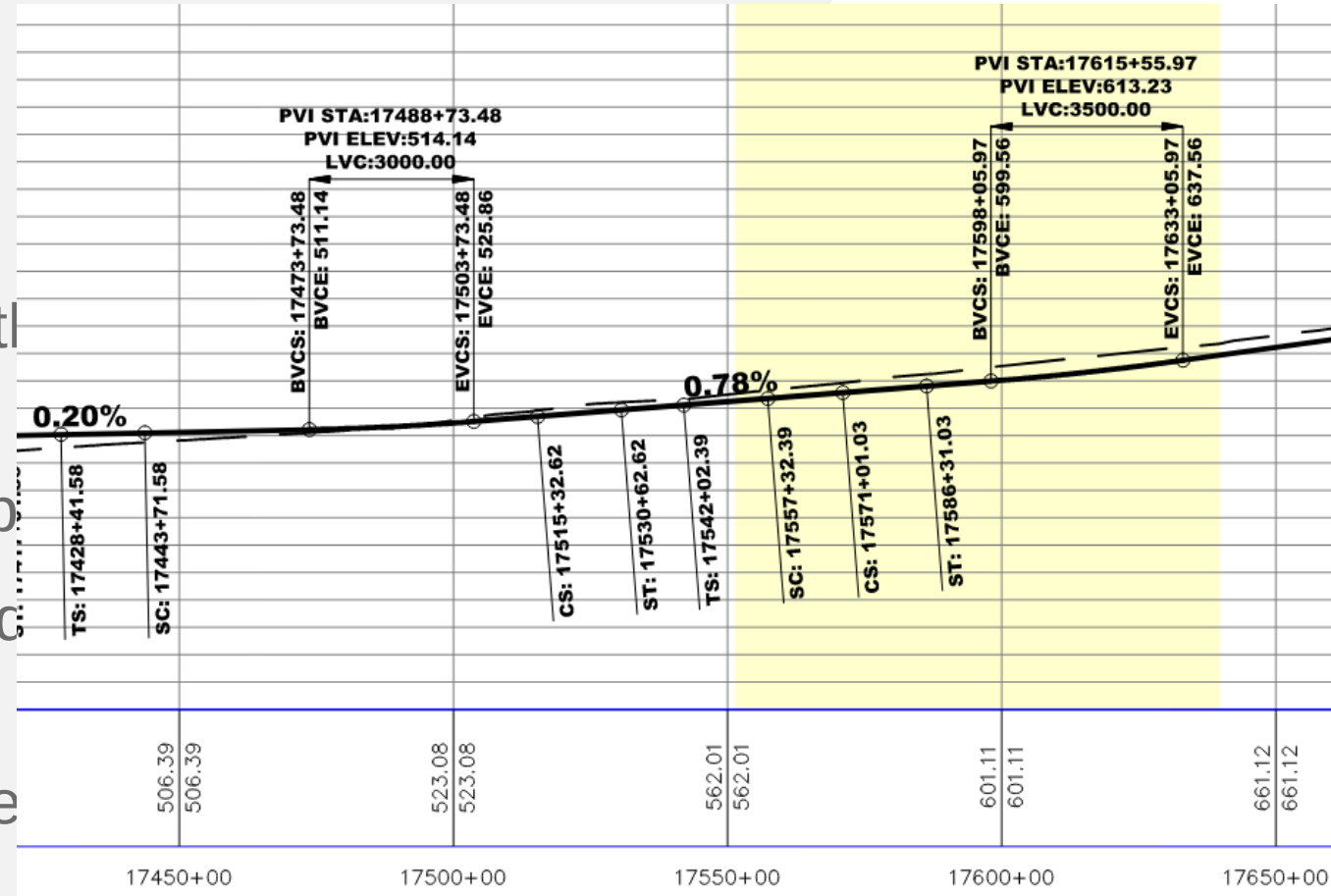
- Expressed in inches of elevation difference between “high rail” (outside of curve) and “low rail” (inside of curve)
- $E = E_a + E_u$
 - E_a – Actual Superelevation
 - E_u – Unbalanced Superelevation (3”-3.5” max)
- Function of design speed and degree of curve
- Max values, E_a :
 - Light Rail: 4”
- Transitioned by spiral



$$E = 4.011 V / R$$

Vertical Design

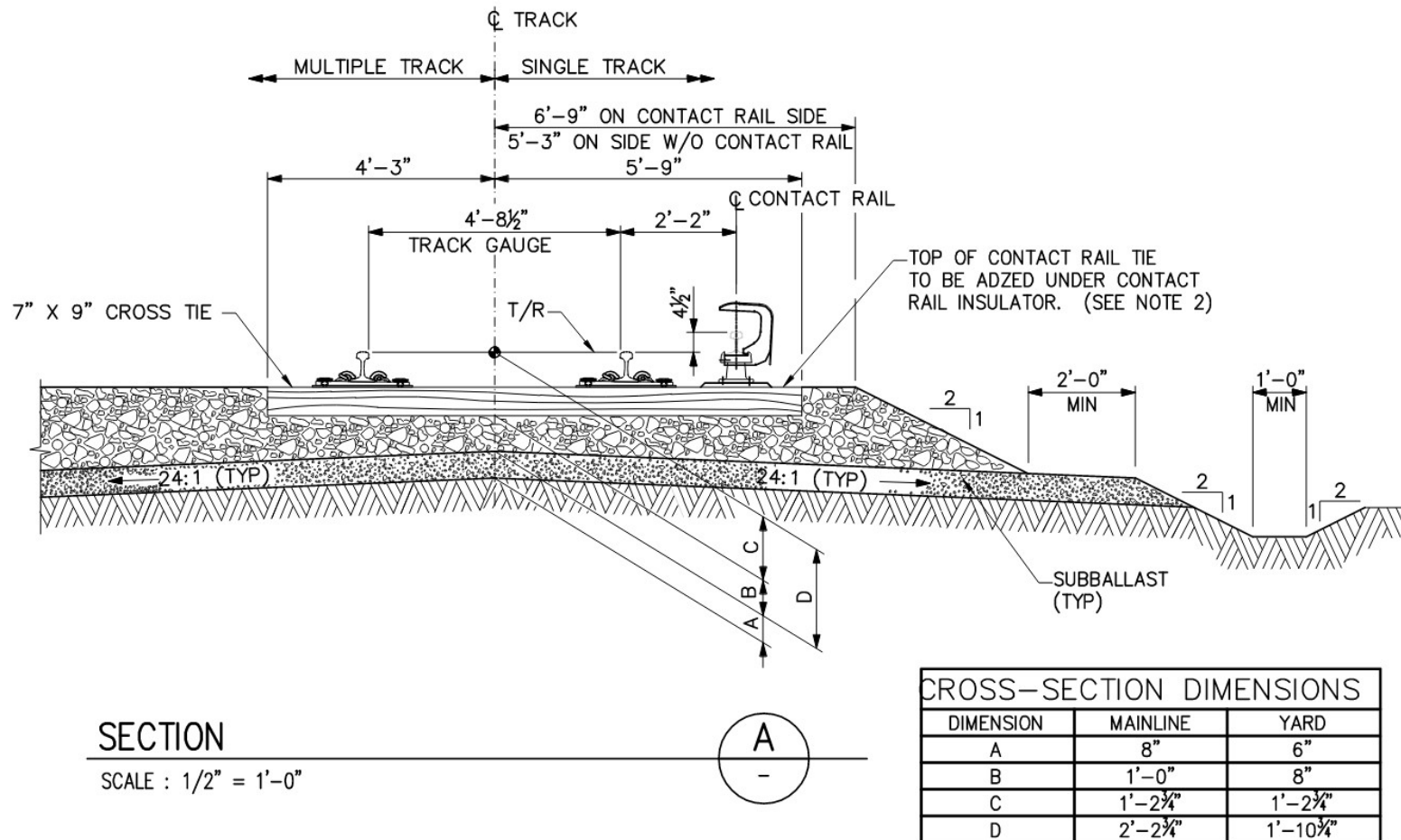
- Mainline track Grade – FLAT
 - Max grade typically 2.0 - 2.5%
- Profile typically shows the top of the low rail in a horizontal curve
- Vertical curves should not overlap
- Vertical curves shall not be placed within turnouts
- Vertical curves preferred not to be placed within horizontal spirals



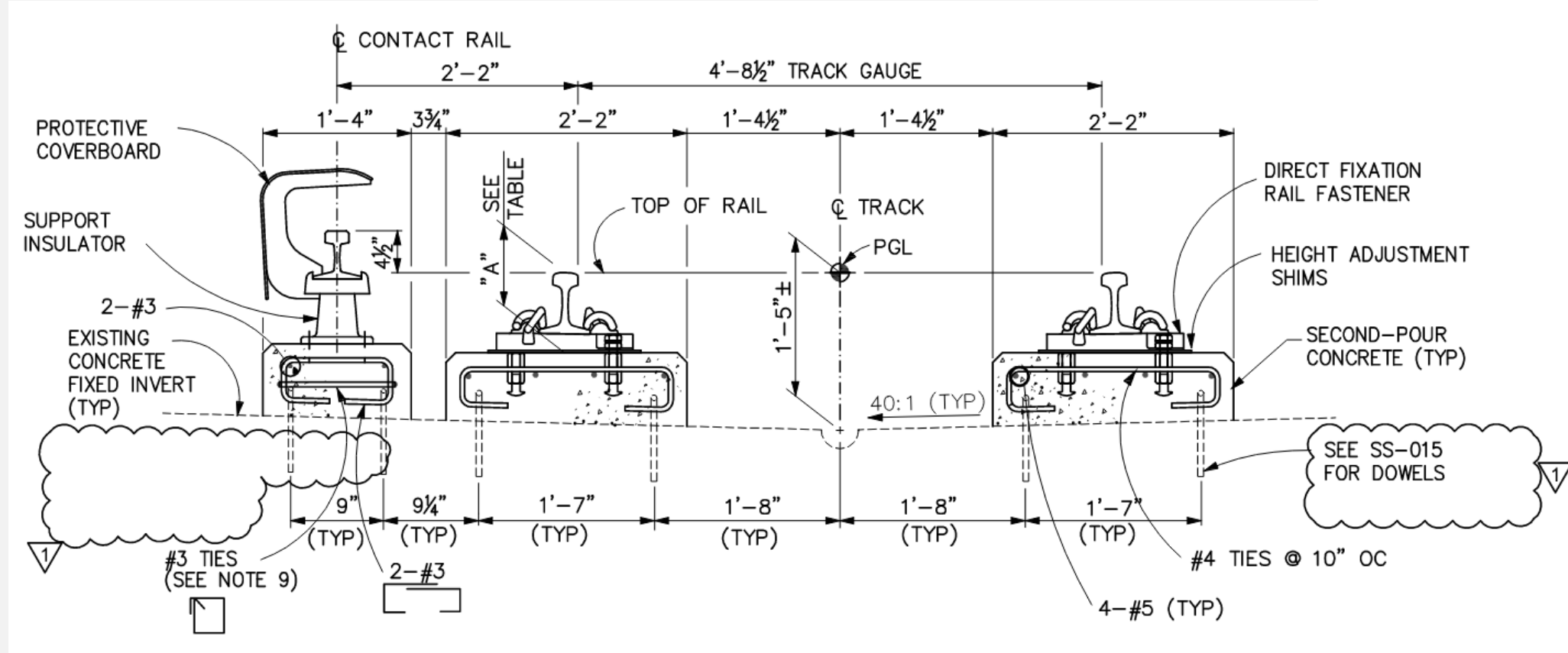


Drainage

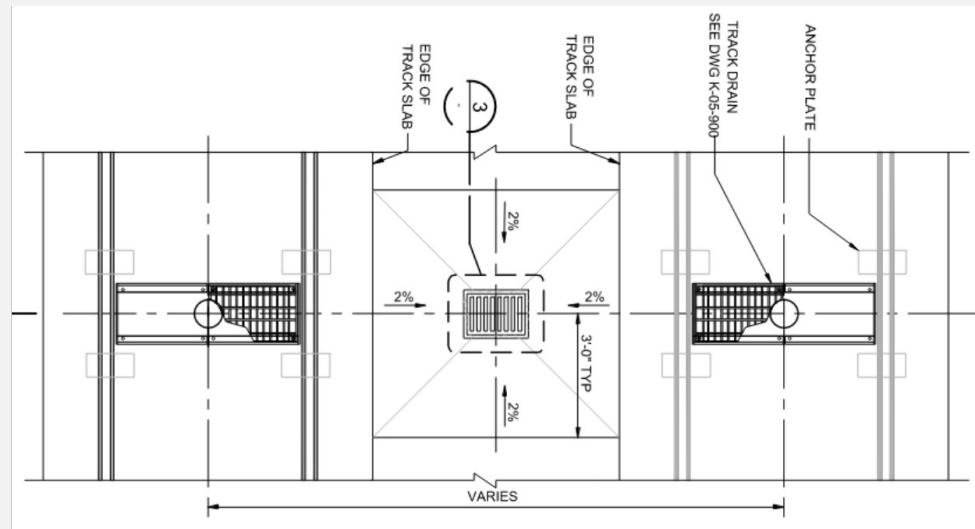
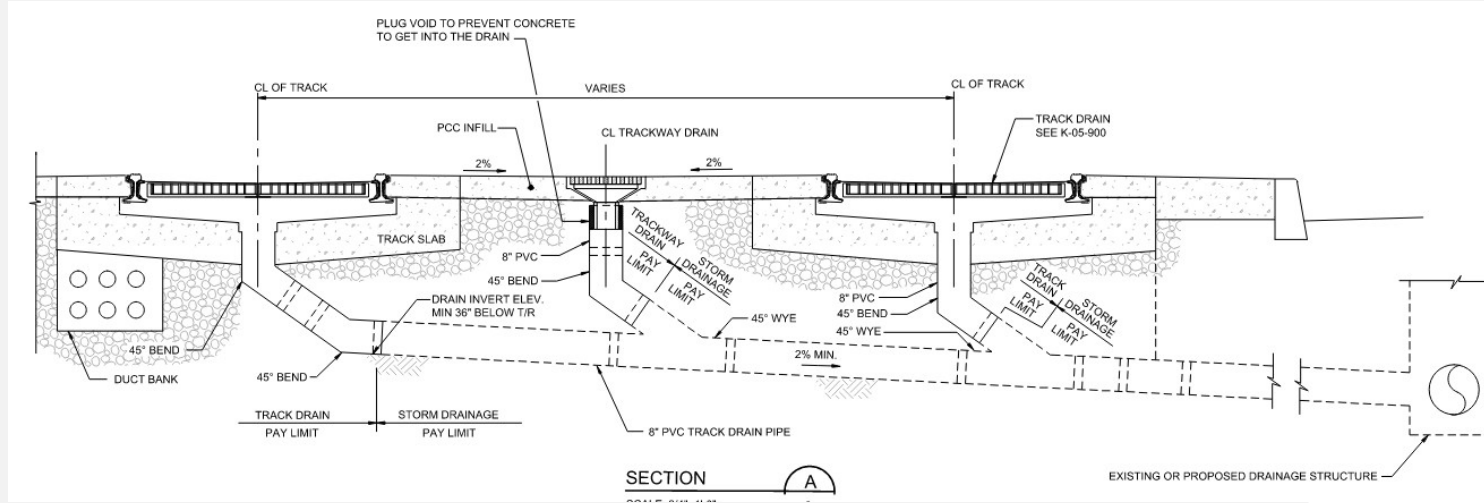
Drainage – Ballasted Track



Drainage – Direct Fixation Track



Drainage – Embedded Track





Track Types

Ballasted Track

Direct Fixation Track

Embedded Track

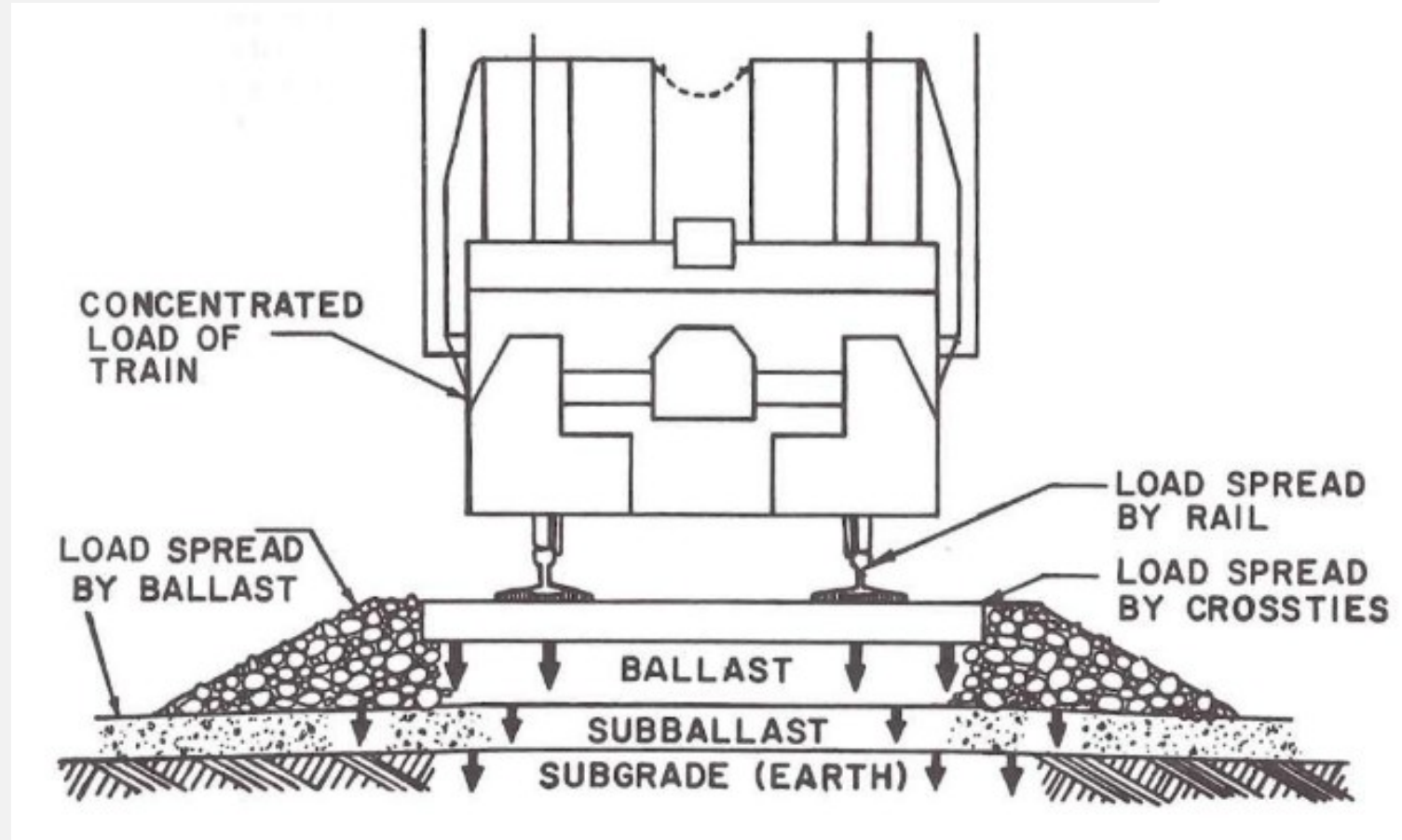
Track Types – Ballasted Track

- Ballasted Track



Track Types - Ballasted Track

- Subgrade
- Subballast
- Ballast
- Cross Ties
- Running Rail
- Drainage



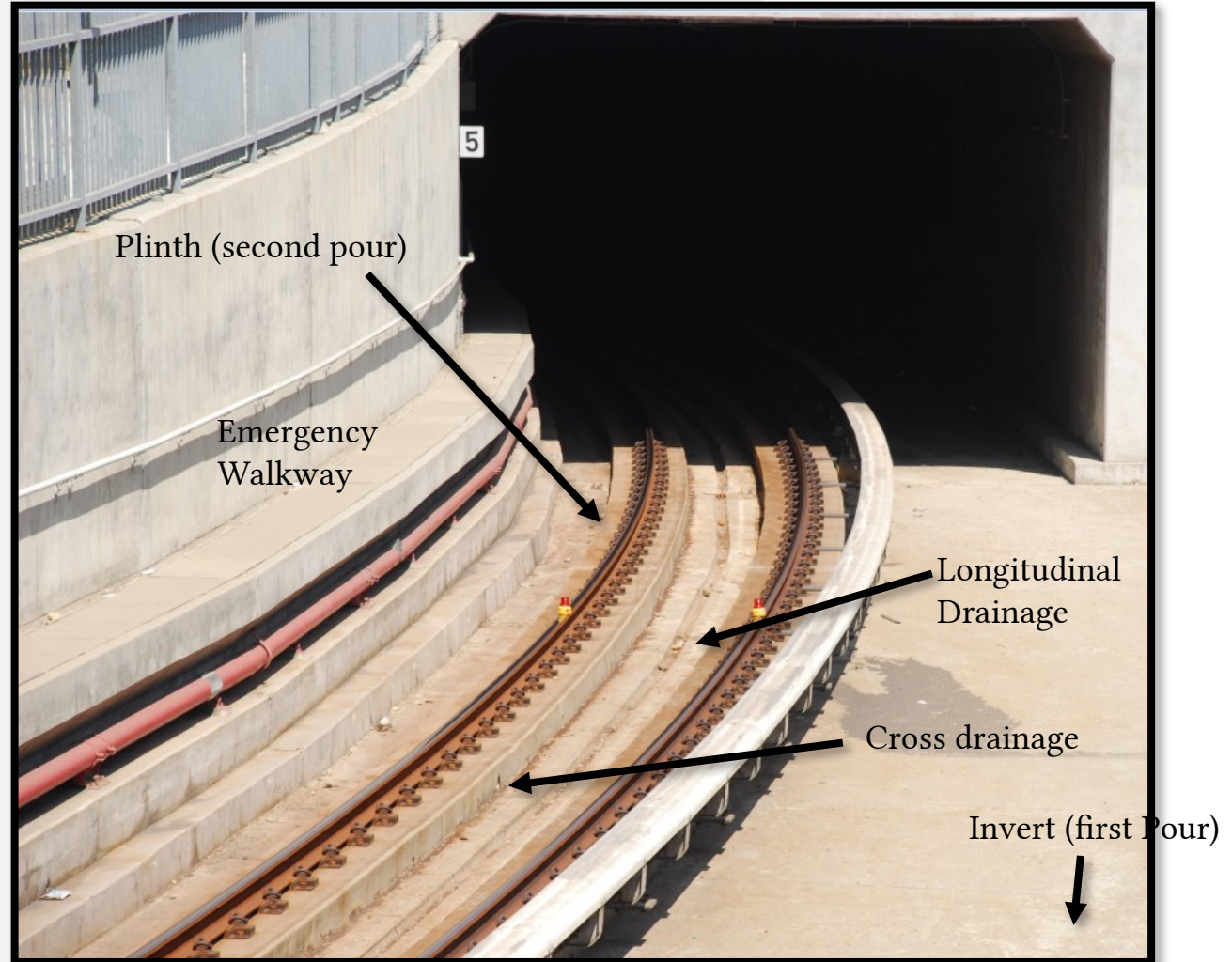
Track Types – Direct Fixation Track

- Direct Fixation Track



Track Types – Direct Fixation Track

- Subgrade
- Track slab (invert)
- Plinths
- DF Fasteners
- Running Rail
- Drainage



Track Types – Direct Fixation Track

- Tighter construction quality control and inspection tolerances for proper installation.
- More cost-effective in the long run with maintenance and use than ballasted track, but more costly up front.

Track Types – Embedded Track

- Embedded Track



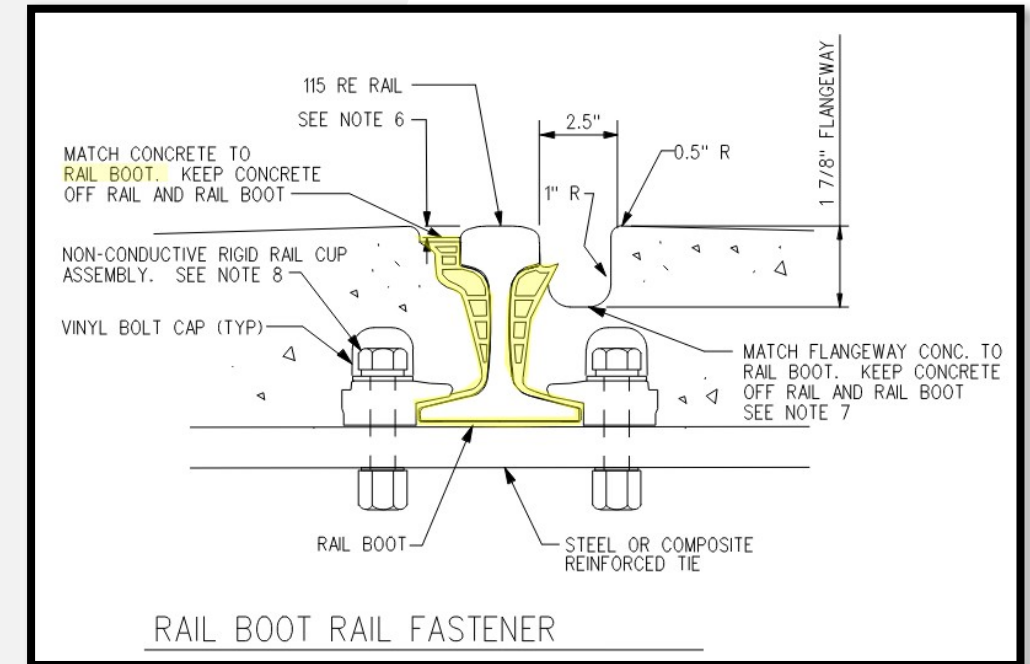
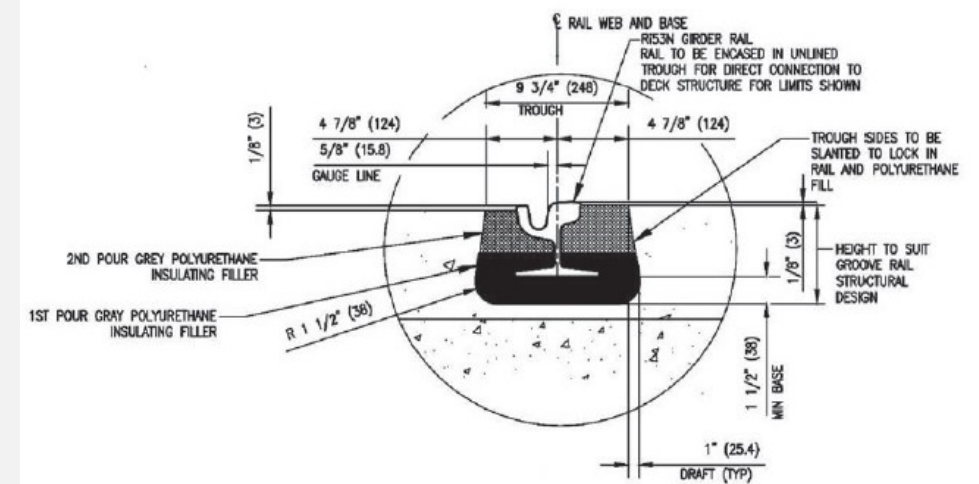
Track Types – Embedded Track

- Subgrade
- Track slab
- Running rail
- Rail insulation
- Drainage



Track Types – Embedded Track

- Light Rail / Street Car



Track Types – Embedded Track

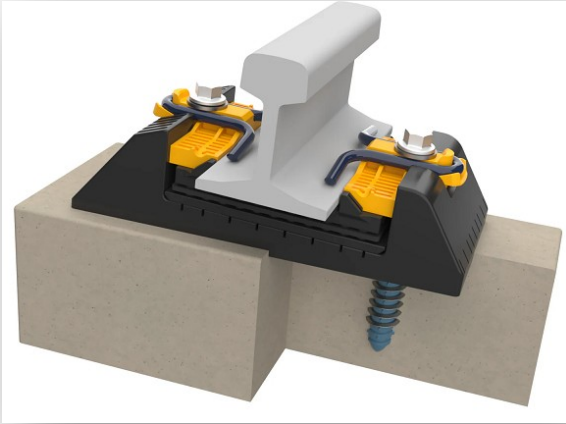
- Design – Rubber Boot over Rail
- Embedded tracks are particularly noisy at special trackwork – turnouts and diamond crossings, where it crosses the flangeway
- Requirements: Vibration mitigation, stray current mitigation



Fastening Systems



Fastening Systems – Elastic Fasteners



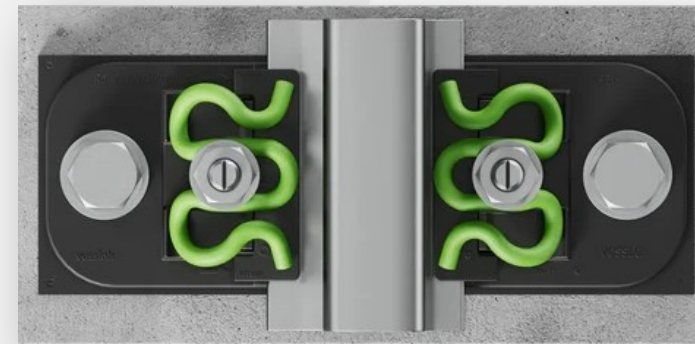
Pandrol - SD DEE Fastener/Basteplate
Credit: Jeff Tuzik



Signum - Nabla Fastening System
Credit: Jeff Tuzik



Voestalpine - Elastic Fastening System SKL
Credit: Jeff Tuzik



Vossloh - 200 Mflex
Credit: Jeff Tuzik



Special Trackwork

Embedded Special Trackwork

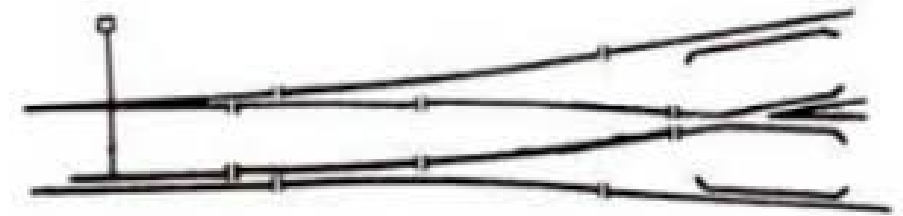


Embedded Special Trackwork

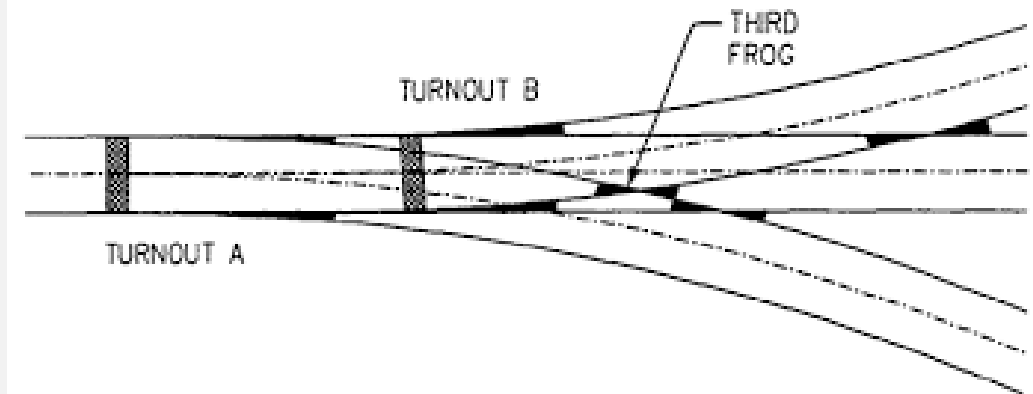


Turnouts

- Lateral Turnout – diverging track to one side
- Equilateral Turnout – diverging tracks arranged equally
 - Benefit - increasing the radius of the guide curve and shortening the length of the turnout
- Lapped Turnout – a second turnout is placed between the switch and the frog of the initial turnout.
 - Benefit - more compact track layout. Typically allows tracks to veer right or left in addition to proceeding straight

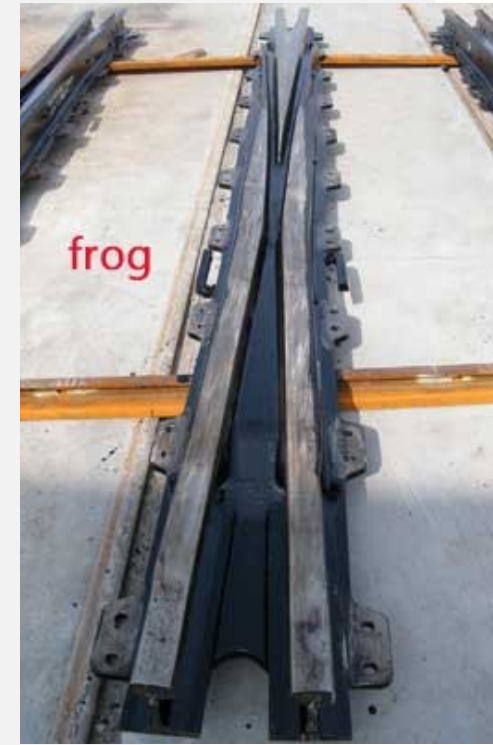


Equilateral Turnout

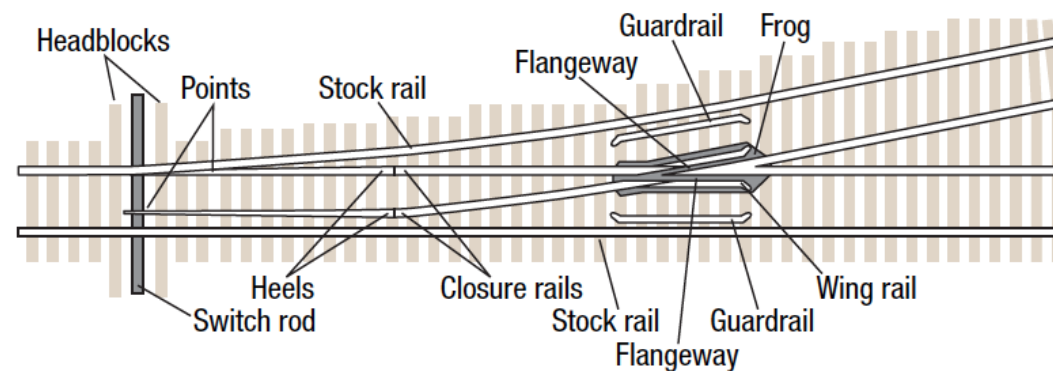


Turnouts

- Allows diverging from one track to another
- Switch, Point of Switch, PTSW or PS
- Identified by “frog number”
 - Higher frog number = higher design speed and gentler offset angle
- Typical turnout numbers:
 - Mainline No. 20 or 24 (Commuter/Freight)
 - Sidings No. 15
 - Yards No 9 and 11 (Commuter/Freight)
 - No 10 and larger (LRT mainline)
 - No 8 or smaller (LRT yard)

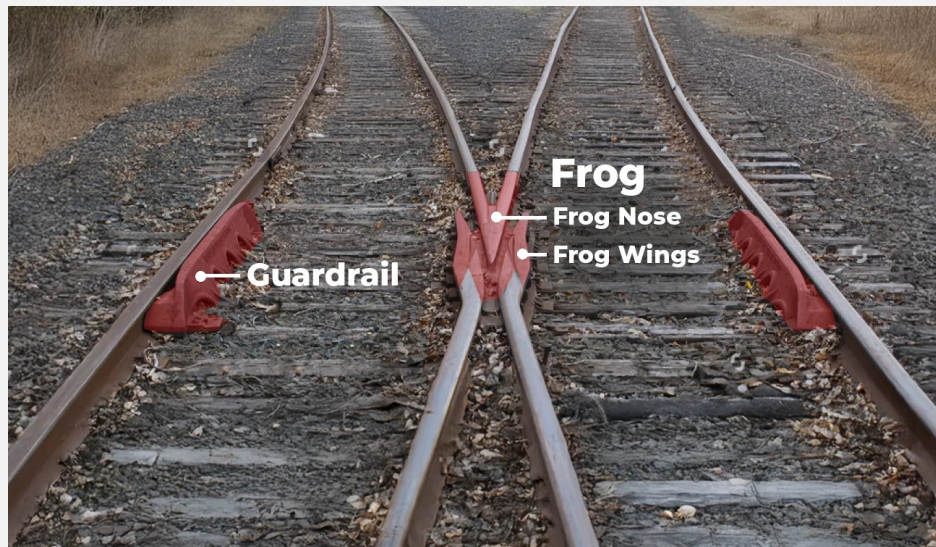


Turnout parts



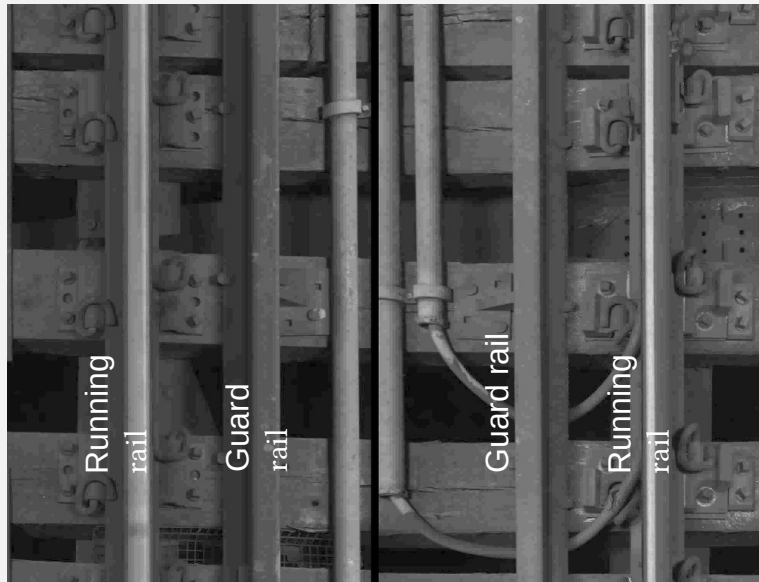
Special Trackwork - Guard Rails

- Guard Rails used in association with frogs. Keeps wheels traveling along the correct pathway in the trough.
- Exception: self-guarded frogs
 - Guarding structure incorporated into the frog structure itself



Special Trackwork - Restraining Rails

- Restraining Rails are used in tight curves. Forces acting on the wheels are not in alignment with the running rail, restraining rails keep it in the correct path.
- Similar type of track used in the UK: Check Rail. It is used extensively there, but check rails are not used widely in the US.



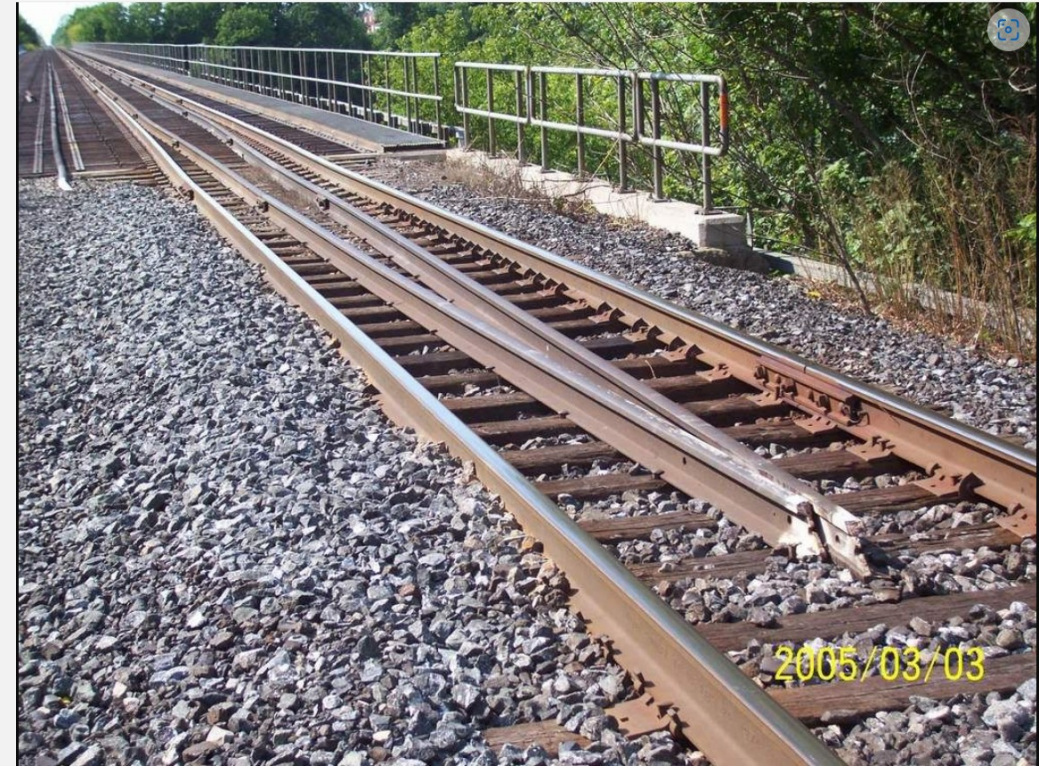
Credit: ARM/NYCT



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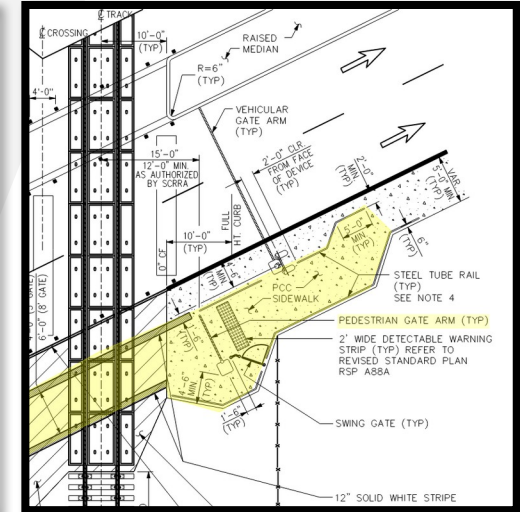
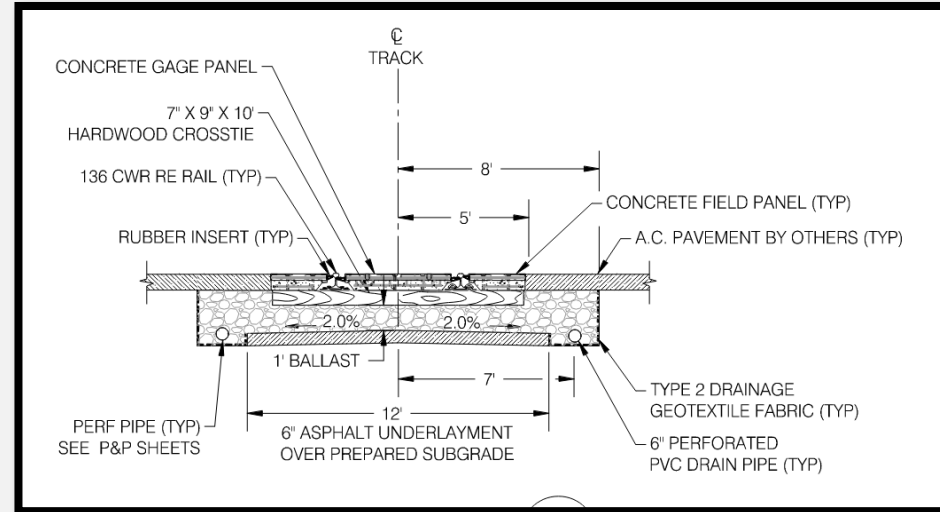
Special Trackwork – Inner Emergency Guard Rail

- Used on track where if a derailment happens, it would be additionally dangerous. Such as a bridge, or if the track runs along a wall or near structures
- The inner emergency guard rail helps steer the train, and reducing the likelihood of a severe derailment

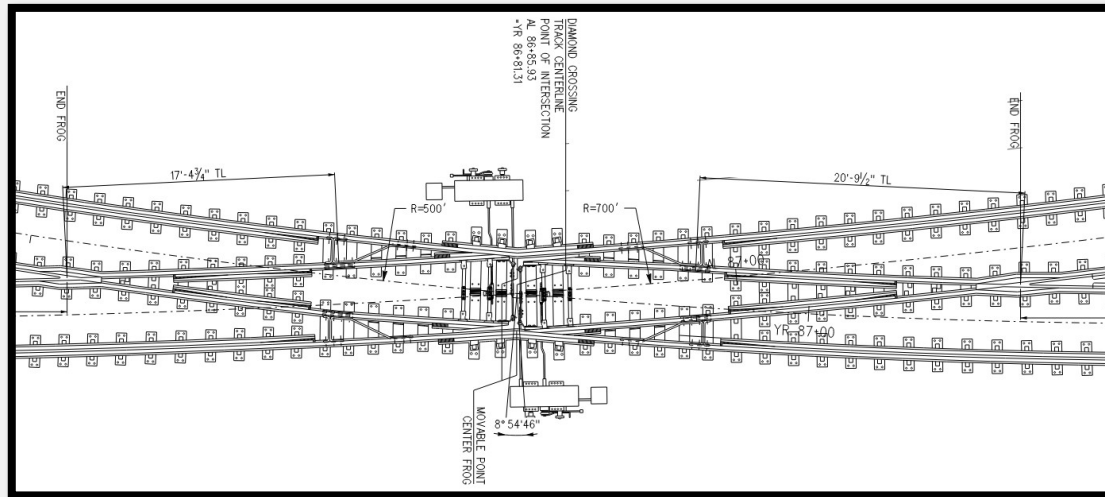


Grade Crossings

- Drainage
- Track
- Crossing Panels
- Roadway
- Gates and Flashers
- Pedestrian Gates



Special Trackwork



MOVABLE POINT CROSSING



DIAMOND CROSSING



DOUBLE SLIP SWITCH



Antique Track

B O S T O

Antique Track



CAMDEN AMBOY RAILROAD
HIGHTSTOWN, NJ



HAYTOR TRAMWAY
HAYTOR, DEVONSHIRE, UK

Antique Track

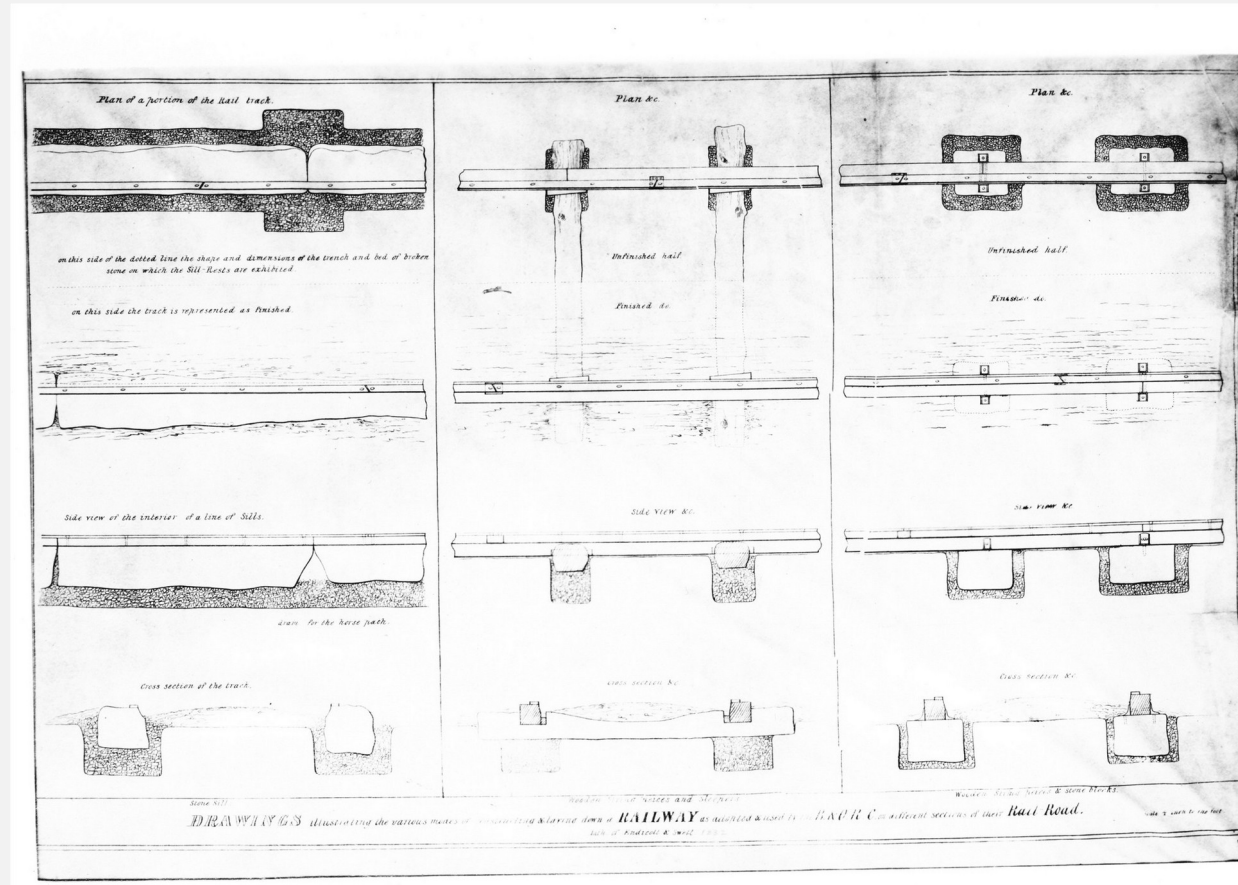


GOLDEN SPIKE NATIONAL PARK
OGDEN, UT



B&O RAILROAD
PATAPSCO STATE PARK
ELLICOTT CITY, MD

Antique Track



B&O RAILROAD SCHEMATICS